

ADV: Calculus (Adv), C3 Applications of Calculus (Adv)
Curve Sketching (Y12)

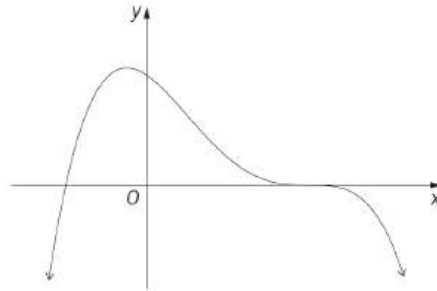
Teacher: Ljiljana Todic

Exam Equivalent Time: 174 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

Questions

1. Calculus, 2ADV C3 2021 HSC 8 MC

The graph of $y = f(x)$ is shown.



Which of the following could be the equation of this graph?

- A. $y = (1 - x)(2 + x)^3$
- B. $y = (x + 1)(x - 2)^3$
- C. $y = (x + 1)(2 - x)^3$
- D. $y = (x - 1)(2 + x)^3$

2. Calculus, 2ADV C3 2011 HSC 7a

Let $f(x) = x^3 - 3x + 2$.

- i. Find the coordinates of the stationary points of $y = f(x)$, and determine their nature. (3 marks)
- ii. Hence, sketch the graph $y = f(x)$ showing all stationary points and the y -intercept. (2 marks)

3. Calculus, 2ADV C3 2005 HSC 4b

A function $f(x)$ is defined by $f(x) = (x + 3)(x^2 - 9)$.

- i. Find all solutions of $f(x) = 0$ (2 marks)
- ii. Find the coordinates of the turning points of the graph of $y = f(x)$, and determine their nature. (3 marks)
- iii. Hence sketch the graph of $y = f(x)$, showing the turning points and the points where the curve meets the x -axis. (2 marks)
- iv. For what values of x is the graph of $y = f(x)$ concave down? (1 mark)

4. Calculus, 2ADV C3 2006 HSC 5a

A function $f(x)$ is defined by $f(x) = 2x^2(3 - x)$.

- i. Find the coordinates of the turning points of $y = f(x)$ and determine their nature. (3 marks)
- ii. Find the coordinates of the point of inflexion. (1 mark)
- iii. Hence sketch the graph of $y = f(x)$, showing the turning points, the point of inflexion and the points where the curve meets the x -axis. (3 marks)
- iv. What is the minimum value of $f(x)$ for $-1 \leq x \leq 4$? (1 mark)

5. Calculus, 2ADV C3 2007 HSC 6b

Let $f(x) = x^4 - 4x^3$.

- i. Find the coordinates of the points where the curve crosses the axes. (2 marks)
- ii. Find the coordinates of the stationary points and determine their nature. (4 marks)
- iii. Find the coordinates of the points of inflexion. (1 mark)
- iv. Sketch the graph of $y = f(x)$, indicating clearly the intercepts, stationary points and points of inflexion. (3 marks)

6. Calculus, 2ADV C3 2015 HSC 13c

Consider the curve $y = x^3 - x^2 - x + 3$.

- i. Find the stationary points and determine their nature. (4 marks)
- ii. Given that the point $P\left(\frac{1}{3}, \frac{70}{27}\right)$ lies on the curve, prove that there is a point of inflexion at P . (2 marks)
- iii. Sketch the curve, labelling the stationary points, point of inflexion and y -intercept. (2 marks)

7. Calculus, 2ADV C3 2016 HSC 13a

Consider the function $y = 4x^3 - x^4$.

- i. Find the two stationary points and determine their nature. (4 marks)
- ii. Sketch the graph of the function, clearly showing the stationary points and the x and y intercepts. (2 marks)

8. Calculus, 2ADV C3 2017 HSC 13b

Consider the curve $y = 2x^3 + 3x^2 - 12x + 7$.

- Find the stationary points of the curve and determine their nature. (4 marks)
 - Sketch the curve, labelling the stationary points. (2 marks)
 - Hence, or otherwise, find the values of x for which $\frac{dy}{dx}$ is positive. (1 mark)
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9. Calculus, 2ADV C3 2018 HSC 13a

Consider the curve $y = 6x^2 - x^3$.

- Find the stationary points and determine their nature. (3 marks)
 - Given that the point (2,16) lies on the curve, show that it is a point of inflexion. (2 marks)
 - Sketch the curve, showing the stationary points, the point of inflexion and the x and y intercepts. (2 marks)
-

10. Calculus, 2ADV C3 2020 HSC 16

Sketch the graph of the curve $y = -x^3 + 3x^2 - 1$, labelling the stationary points and point of inflection. Do NOT determine the x -intercepts of the curve. (4 marks)

11. Calculus, 2ADV C3 2019 HSC 14b

The derivative of a function $y = f(x)$ is given by $f'(x) = 3x^2 + 2x - 1$.

- Find the x -values of the two stationary points of $y = f(x)$, and determine the nature of the stationary points. (2 marks)
 - The curve passes through the point (0, 4).
Find an expression for $f(x)$. (2 marks)
 - Hence sketch the curve, clearly indicating the stationary points. (2 marks)
 - For what values of x is the curve concave down? (1 mark)
-

12. Calculus, 2ADV C3 2004 HSC 4b

Consider the function $f(x) = x^3 - 3x^2$.

- Find the coordinates of the stationary points of the curve $y = f(x)$ and determine their nature. (3 marks)
 - Sketch the curve showing where it meets the axes. (2 marks)
 - Find the values of x for which the curve $y = f(x)$ is concave up. (2 marks)
-

13. Calculus, 2ADV C3 2010 HSC 6a

Let $f(x) = (x + 2)(x^2 + 4)$.

- Show that the graph $y = f(x)$ has no stationary points. (2 marks)
 - Find the values of x for which the graph $y = f(x)$ is concave down, and the values for which it is concave up. (2 marks)
 - Sketch the graph $y = f(x)$, indicating the values of the x and y intercepts. (2 marks)
-

14. Calculus, 2ADV C3 2008 HSC 8a

Let $f(x) = x^4 - 8x^2$.

- Find the coordinates of the points where the graph of $y = f(x)$ crosses the axes. (2 marks)
 - Show that $f(x)$ is an even function. (1 mark)
 - Find the coordinates of the stationary points of $f(x)$ and determine their nature. (4 marks)
 - Sketch the graph of $y = f(x)$. (1 mark)
-

15. Calculus, 2ADV C3 2004 HSC 9c

Consider the function $f(x) = \frac{\log_e x}{x}$, for $x > 0$.

- Show that the graph of $y = f(x)$ has a stationary point at $x = e$. (2 marks)
 - By considering the gradient on either side of $x = e$, or otherwise, show that the stationary point is a maximum. (1 mark)
 - Use the fact that the maximum value of $f(x)$ occurs at $x = e$ to deduce that $e^x \geq x^e$ for all $x > 0$. (2 marks)
-

16. Calculus, 2ADV C3 2009 HSC 10

Let $f(x) = x - \frac{x^2}{2} + \frac{x^3}{3}$

a. Show that the graph of $y = f(x)$ has no turning points. (2 marks)

b. Find the point of inflexion of $y = f(x)$. (1 mark)

c. i. Show that $1 - x + x^2 - \frac{1}{1+x} = \frac{x^3}{1+x}$ for $x \neq -1$. (1 mark)

ii. Let $g(x) = \ln(1+x)$.

Use the result in part c.i. to show that $f'(x) \geq g'(x)$ for all $x \geq 0$. (2 marks)

d. On the same set of axes, sketch the graphs of $y = f(x)$ and $y = g(x)$ for $x \geq 0$. (2 marks)

e. Show that $\frac{d}{dx}[(1+x)\ln(1+x) - (1+x)] = \ln(1+x)$. (2 marks)

f. Find the area enclosed by the graphs of $y = f(x)$ and $y = g(x)$, and the straight line $x = 1$. (2 marks)

17. Calculus, 2ADV C3 2012 HSC 14a

A function is given by $f(x) = 3x^4 + 4x^3 - 12x^2$.

i. Find the coordinates of the stationary points of $f(x)$ and determine their nature. (3 marks)

ii. Hence, sketch the graph $y = f(x)$ showing the stationary points. (2 marks)

iii. For what values of x is the function increasing? (1 mark)

iv. For what values of k will $f(x) = 3x^4 + 4x^3 - 12x^2 + k = 0$ have no solution? (1 mark)

Worked Solutions

1. Calculus, 2ADV C3 2021 HSC 8 MC

By elimination:

A single negative root occurs when $x = -1$

→ Eliminate A and D

When $x = 0$, $y > 0$

→ Eliminate B

⇒ C

2. Calculus, 2ADV C3 2011 HSC 7a

i. $f(x) = x^3 - 3x + 2$

$$f'(x) = 3x^2 - 3$$

$$f''(x) = 6x$$

Stationary points when $f'(x) = 0$

$$3x^2 - 3 = 0$$

$$3(x^2 - 1) = 0$$

$$\therefore x^2 = 1$$

$$x = \pm 1$$

When $x = 1$

$$f(1) = 1 - 3 + 2 = 0$$

$$f''(1) = 6 > 0$$

∴ MIN S.P. at $(1, 0)$

When $x = -1$

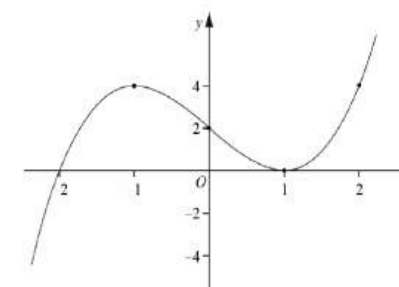
$$f(-1) = -1 + 3 + 2 = 4$$

$$f''(-1) = -6 < 0$$

∴ MAX S.P. at $(-1, 4)$

ii. $y = x^3 - 3x + 2$

$$y\text{-intercept} = 2$$



MARKER'S COMMENT: Graphs should be large (around $\frac{1}{2}$ page), axes drawn with a ruler, with intercepts and turning points clearly shown. The scale can be different on each axis for clarity, as shown in the Worked Solution.

3. Calculus, 2ADV C3 2005 HSC 4b

i. $f(x) = (x+3)(x^2-9)$
 $= (x+3)(x+3)(x-3)$
 $\therefore f(x) = 0$ when $x = -3$ or 3

ii. $f(x) = (x+3)(x^2-9)$
 $= x^3 - 9x + 3x^2 - 27$
 $= x^3 + 3x^2 - 9x - 27$
 $f'(x) = 3x^2 + 6x - 9$
 $f''(x) = 6x + 6$

S.P.'s when $f'(x) = 0$

$$3x^2 + 6x - 9 = 0$$

$$3(x^2 + 2x - 3) = 0$$

$$3(x-1)(x+3) = 0$$

At $x = 1$

$$f(1) = (4)(-8) = -32$$

$$f''(1) = 6 + 6 = 12 > 0$$

$\therefore$ MIN at $(1, -32)$

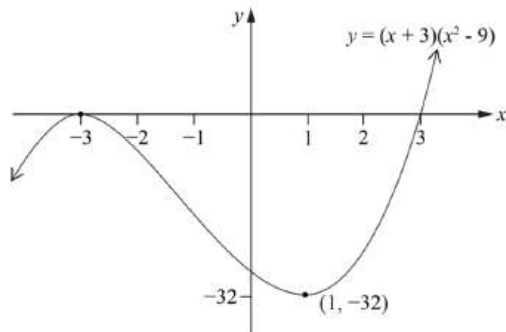
At $x = -3$

$$f(-3) = 0$$

$$f''(-3) = (6 \times -3) + 6 = -12 < 0$$

$\therefore$ MAX at $(-3, 0)$

iii.



iv. $f(x)$ is concave down when

$$f''(x) < 0$$

$$6x + 6 < 0$$

$$6x < -6$$

$$x < -1$$

4. Calculus, 2ADV C3 2006 HSC 5a

i. $f(x) = 2x^2(3-x)$
 $= 6x^2 - 2x^3$
 $f'(x) = 12x - 6x^2$
 $f''(x) = 12 - 12x$

S.P.'s when $f'(x) = 0$

$$12x - 6x^2 = 0$$

$$6x(2-x) = 0$$

$$x = 0 \text{ or } 2$$

When $x = 0$

$$f(0) = 0$$

$$f''(0) = 12 - 0 = 12 > 0$$

$\therefore$ MIN at $(0, 0)$

When $x = 2$

$$f(2) = 2 \times 2^2(3-2) = 8$$

$$f''(2) = 12 - (12 \times 2) = -12 < 0$$

$\therefore$ MAX at $(2, 8)$

ii. P.I. when $f''(x) = 0$

$$12 - 12x = 0$$

$$12x = 12$$

$$x = 1$$

$$f''(0.5) = 6 > 0$$

$$f''(1.5) = -6 < 0$$

Since concavity changes $\Rightarrow$ P.I. exists

$$f(1) = 2 \times 1^2(3-1)$$

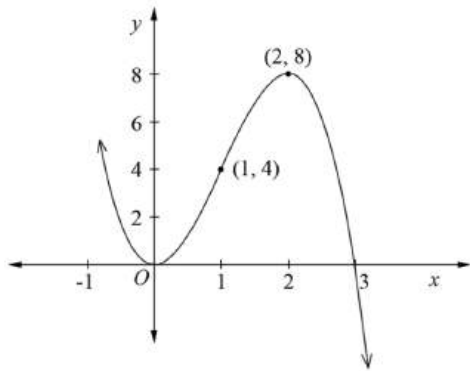
$$= 4$$

$\therefore$ P.I. at $(1, 4)$

iii. $f(x)$ meets x -axis when $f(x) = 0$

$$2x^2 \times (3-x) = 0$$

$$x = 0 \text{ or } 3$$



- (iv) The graph clearly shows that in the given range $-1 \leq x \leq 4$, the minimum will occur when $x = 4$

$$\begin{aligned}\therefore \text{Minimum} &= 2 \times 4^2(3 - 4) \\ &= -32\end{aligned}$$

5. Calculus, 2ADV C3 2007 HSC 6b

i. $f(x) = x^4 - 4x^3$

Cuts x -axis when $f(x) = 0$

$$x^4 - 4x = 0$$

$$x^3(x - 4) = 0$$

$$x = 0 \text{ or } 4$$

$\therefore$ Cuts the x -axis at $(0, 0)$, $(4, 0)$

Cuts the y -axis when $x = 0$

$\therefore$ Cuts the y -axis at $(0, 0)$

ii. $f(x) = x^4 - 4x^3$

$$f'(x) = 4x^3 - 12x^2$$

$$f''(x) = 12x^2 - 24x$$

S.P.'s when $f'(x) = 0$

$$4x^3 - 12x^2 = 0$$

$$4x^2(x - 3) = 0$$

$$x = 0 \text{ or } 3$$

When $x = 0$

$$f(0) = 0$$

$$f''(0) = 0$$

x	-0.5	0	0.5
$f''(x)$	> 0	0	< 0

Since concavity changes, a P.I. occurs at $(0, 0)$

When $x = 3$

$$f(3) = 3^4 - 4 \times 3^3$$

$$= -27$$

$$f''(3) = 12 \times 3^2 - 24 \times 3$$

$$= 36 > 0$$

$\therefore$ Minimum S.P. at $(3, -27)$

iii. P.I. when $f''(x) = 0$

$$12x^2 - 24x = 0$$

$$12x(x - 2) = 0$$

$$x = 0 \text{ or } 2$$

P.I. at $(0, 0)$ (from(ii))

When $x = 2$

x	1.5	2	2.5
$f''(x)$	< 0	0	> 0

Since concavity changes, a P.I.

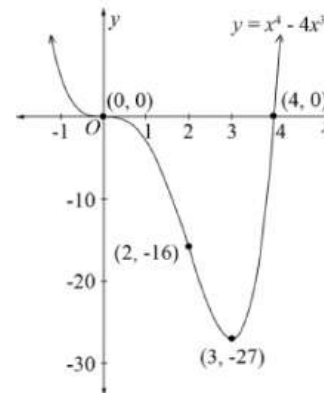
occurs when $x = 2$

$$f(2) = 2^4 - 4 \times 2^3$$

$$= 16$$

$\therefore$ P.I.'s at $(0, 0)$ and $(2, 16)$

iv.



6. Calculus, 2ADV C3 2015 HSC 13c

i. $y = x^3 - x^2 - x + 3$

$$\frac{dy}{dx} = 3x^2 - 2x - 1$$

$$\frac{d^2y}{dx^2} = 6x - 2$$

S.P.'s when $\frac{dy}{dx} = 0$

$$3x^2 - 2x - 1 = 0$$

$$(3x + 1)(x - 1) = 0$$

$$x = -\frac{1}{3} \text{ or } 1$$

When $x = -\frac{1}{3}$

$$\begin{aligned} f\left(-\frac{1}{3}\right) &= \left(-\frac{1}{3}\right)^3 - \left(-\frac{1}{3}\right)^2 - \left(-\frac{1}{3}\right) + 3 \\ &= -\frac{1}{27} - \frac{1}{9} + \frac{1}{3} + 3 \\ &= \frac{86}{27} \end{aligned}$$

$$f''\left(-\frac{1}{3}\right) = \left(6 \times -\frac{1}{3}\right) - 2 = -4 < 0$$

$$\therefore \text{MAX at } \left(-\frac{1}{3}, \frac{86}{27}\right)$$

When $x = 1$

$$f(1) = 1^3 - 1^2 - 1 + 3 = 2$$

$$f''(1) = (6 \times 1) - 2 = 4 > 0$$

$$\therefore \text{MIN at } (1, 2)$$

ii. $\frac{d^2y}{dx^2} = 0$ when

$$6x - 2 = 0$$

$$x = \frac{1}{3}$$

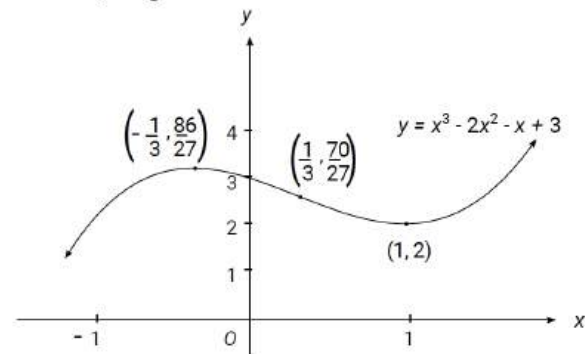
Checking change of concavity

x	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$
$f(x)$	$-1 < 0$	0	$1 > 0$

Concavity changes either side of $x = \frac{1}{3}$ $\therefore \left(\frac{1}{3}, \frac{70}{27}\right)$ is a P.I.

iii. When $x = 0$

$$y = 3$$



7. Calculus, 2ADV C3 2016 HSC 13a

i. $y = 4x^3 - x^4$

$$y' = 12x^2 - 4x^3$$

$$y'' = 24x - 12x^2$$

S.P.'s when $y' = 0$,

$$12x^2 - 4x^3 = 0$$

$$4x^2(3 - x) = 0$$

$$\therefore x = 0 \text{ or } 3$$

When $x = 0$, $y''(0) = 0$

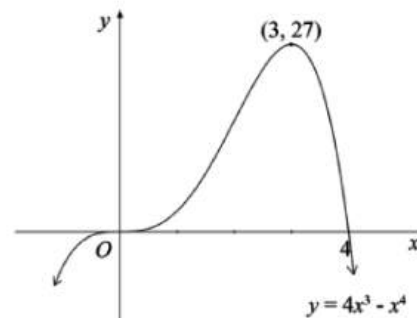
$$\therefore \text{P.I. at } (0, 0)$$

When $x = 3$,

$$y''(3) = 24(3) - 12(9) = -36 < 0$$

$$\therefore \text{MAX at } (3, 27)$$

ii.



8. Calculus, 2ADV C3 2017 HSC 13b

i. $y = 2x^3 + 3x^2 - 12x + 7$

$$\frac{dy}{dx} = 6x^2 + 6x - 12$$

$$\frac{d^2y}{dx^2} = 12x + 6$$

S.P. when $\frac{dy}{dx} = 0$

$$6x^2 + 6x - 12 = 0$$

$$x^2 + x - 2 = 0$$

$$(x + 2)(x - 1) = 0$$

$$x = -2 \text{ or } 1$$

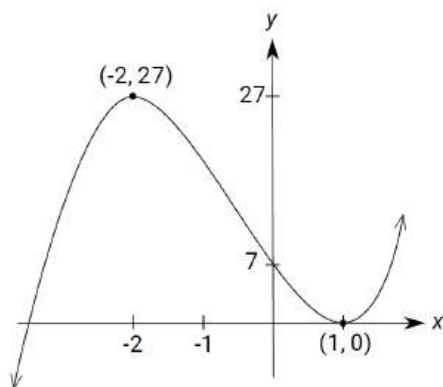
When $x = -2$, $\frac{d^2y}{dx^2} < 0$

$$\therefore \text{MAX at } (-2, 27)$$

When $x = 1$, $\frac{d^2y}{dx^2} > 0$

$$\therefore \text{MIN at } (1, 0)$$

ii.



iii. Solution 1

From graph, gradient is positive for

$$x < -2 \text{ and } x > 1$$

$$\therefore \frac{dy}{dx} > 0 \text{ for } x < -2 \text{ and } x > 1$$

Solution 2

$$\frac{dy}{dx} > 0$$

$$6x^2 + 6x - 12 > 0$$

$$(x + 2)(x - 1) > 0$$



$$\therefore x < -2 \text{ and } x > 1$$

9. Calculus, 2ADV C3 2018 HSC 13a

i. $y = 6x^2 - x^3$

$$\frac{dy}{dx} = 12x - 3x^2$$

$$\frac{d^2y}{dx^2} = 12 - 6x$$

S.P.s occur when $\frac{dy}{dx} = 0$

$$12x - 3x^2 = 0$$

$$3x(4 - x) = 0$$

$$x = 0 \text{ or } 4$$

When $x = 0$, $\frac{d^2y}{dx^2} > 0$

$$\therefore \text{MIN at } (0, 0)$$

When $x = 4$, $\frac{d^2y}{dx^2} < 0$

$$\therefore \text{MAX at } (4, 32)$$

ii. P.I. occur when $\frac{d^2y}{dx^2} = 0$,

$$12 - 6x = 0$$

$$x = 2$$

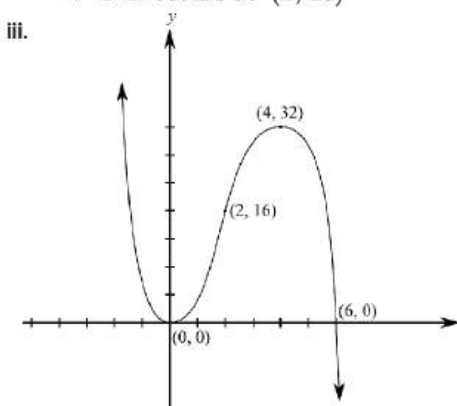
When $x = 2$, $y = 16$

x	2^-	2	2^+
$\frac{d^2y}{dx^2}$	> 0	0	< 0

Since the concavity changes

$$\Rightarrow \text{P.I. occurs at } (2, 16)$$

iii.



10. Calculus, 2ADV C3 2020 HSC 16

$$y = -x^3 + 3x^2 - 1$$

$$\frac{dy}{dx} = -3x^2 + 6x$$

$$\frac{d^2y}{dx^2} = -6x + 6$$

$$\text{SP's when } \frac{dy}{dx} = 0$$

$$-3x^2 + 6x = 0$$

$$-3x(x - 2) = 0$$

$$x = 0 \text{ or } 2$$

When $x = 0$,

$$y = -1$$

$$\frac{d^2y}{dx^2} = 6 > 0$$

$\therefore$ MIN at $(0, -1)$

When $x = 2$,

$$y = -8 + 12 - 1 = 3$$

$$\frac{d^2y}{dx^2} = -6 \times 2 + 6 = -6 < 0$$

$\therefore$ MAX at $(2, 3)$

$$\frac{d^2y}{dx^2} = 0 \text{ when}$$

$$-6x + 6 = 0$$

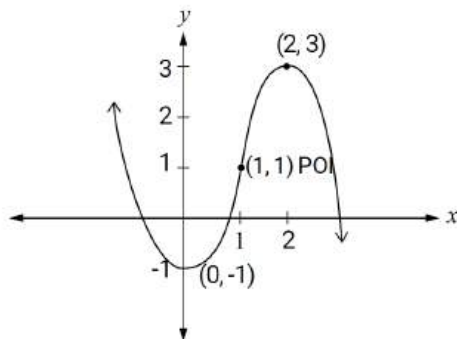
$$x = 1$$

Checking change of concavity

x	$\frac{1}{2}$	1	$\frac{3}{2}$
$\frac{d^2y}{dx^2}$	3	0	-3

Concavity changes either side of $x = 1$

$\therefore$ POI at $(1, 1)$



11. Calculus, 2ADV C3 2019 HSC 14b

$$\text{a. } f'(x) = 3x^2 + 2x - 1$$

$$f''(x) = 6x + 2$$

S.P.'s when $f'(x) = 0$

$$3x^2 + 2x - 1 = 0$$

$$(3x - 1)(x + 1) = 0$$

$$x = \frac{1}{3} \text{ or } -1$$

When $x = \frac{1}{3}$,

$$f''(x) = 4 > 0 \Rightarrow \text{MIN}$$

When $x = -1$,

$$f''(x) = -4 < 0 \Rightarrow \text{MAX}$$

$$\text{b. } f(x) = \int f'(x) dx$$

$$= \int 3x^2 + 2x - 1 dx$$

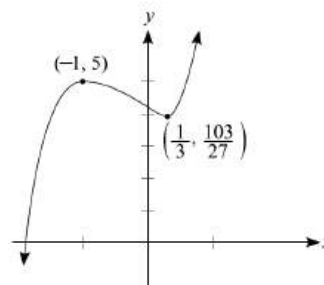
$$= x^3 + x^2 - x + c$$

$$(0, 4) \text{ lies on } f(x) \Rightarrow c = 4$$

$$\therefore f(x) = x^3 + x^2 - x + 4$$

c. When $x = -1$, $y = 5$

$$\text{When } x = \frac{1}{3}, y = \frac{103}{27}$$



d. Concave down when $f''(x) < 0$

$$6x + 2 < 0$$

$$6x < -2$$

$$x < -\frac{1}{3}$$

12. Calculus, 2ADV C3 2004 HSC 4b

(i) $f(x) = x^3 - 3x^2$

$$f'(x) = 3x^2 - 6x$$

$$f''(x) = 6x - 6$$

S.P.'s when $f'(x) = 0$

$$3x^2 - 6x = 0$$

$$3x(x - 2) = 0$$

$$x = 0 \text{ or } 2$$

When $x = 0$

$$f(0) = 0$$

$$f''(0) = 0 - 6 = -6 < 0$$

$\therefore$ MAX at (0, 0)

When $x = 2$

$$f(2) = 2^3 - (3 \times 4) = -4$$

$$f''(2) = (6 \times 2) - 6 = 6 > 0$$

$\therefore$ MIN at (2, -4)

(ii) $f(x) = x^3 - 3x^2$ meets the x -axis when $f(x) = 0$

$$x^3 - 3x^2 = 0$$

$$x^2(x - 3) = 0$$

$$x = 0 \text{ or } 3$$

(iii) $f(x)$ is concave up when

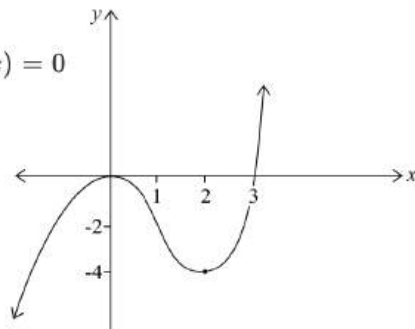
$$f''(x) > 0$$

$$6x - 6 > 0$$

$$6x > 6$$

$$x > 1$$

$\therefore f(x)$ is concave up when $x > 1$



13. Calculus, 2ADV C3 2010 HSC 6a

i. Need to show no S.P.'s

$$f(x) = (x + 2)(x^2 + 4)$$

$$= x^3 + 2x^2 + 4x + 8$$

$$f'(x) = 3x^2 + 4x + 4$$

S.P.s occur when $f'(x) = 0$,

$$3x^2 + 4x + 4 = 0$$

$$\Delta = b^2 - 4ac$$

$$= 4^2 - (4 \times 3 \times 4)$$

$$= 16 - 48$$

$$= -32 < 0$$

Since $\Delta < 0$, No Solution

$\therefore$ No S.P.'s for $f(x)$

ii. $f(x)$ is concave down when $f''(x) < 0$

$$f''(x) = 6x + 4$$

$$\Rightarrow 6x + 4 < 0$$

$$6x < -4$$

$$x < -\frac{2}{3}$$

$\therefore f(x)$ is concave down when $x < -\frac{2}{3}$

$f(x)$ is concave up when $f''(x) > 0$

$$f''(x) = 6x + 4$$

$$\Rightarrow 6x + 4 > 0$$

$$6x > -4$$

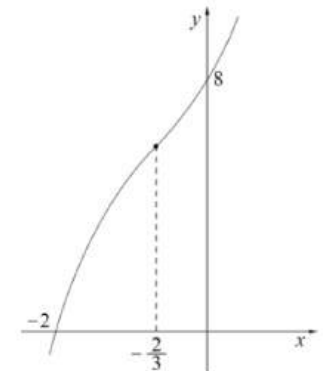
$$x > -\frac{2}{3}$$

$\therefore f(x)$ is concave up when $x > -\frac{2}{3}$

iii. y -intercept $= 2 \times 4 = 8$

x -intercept $= -2$

MARKER'S COMMENT: The significance of the sign of the second derivative was not well understood by most students.



♦♦ Mean mark 33%.
MARKER'S COMMENT: Students are reminded to bring a ruler to the

14. Calculus, 2ADV C3 2008 HSC 8a

i. $f(x) = x^4 - 8x^2$

y -intercept when $x = 0$

$\therefore$ Cuts y axis at $(0, 0)$

x -intercept when $f(x) = 0$

$$x^4 - 8x^2 = 0$$

$$x^2(x^2 - 8) = 0$$

$$x^2 - 8 = 0 \quad \text{or} \quad x = 0$$

$$x^2 = 8$$

$$x = \pm \sqrt{8} = \pm 2\sqrt{2}$$

$\therefore$ Cuts x -axis at $(-2\sqrt{2}, 0)$, and $(2\sqrt{2}, 0)$

(Note that it only touches x -axis at $(0, 0)$)

ii. $f(x) = x^4 - 8x^2$

$$f(-x) = (-x)^4 - 8(-x)^2$$

$$= x^4 - 8x^2$$

$$= f(x)$$

$\therefore f(x)$ is an even function.

iii. $f(x) = x^4 - 8x^2$

$$f'(x) = 4x^3 - 16x$$

$$f''(x) = 12x^2 - 16$$

S.P. when $f'(x) = 0$

$$4x^3 - 16x = 0$$

$$4x(x^2 - 4) = 0$$

$$x^2 - 4 = 0 \quad x = 0$$

$$x^2 = 4$$

$$x = \pm 2$$

$$\text{At } x = 2 \quad f(x) = (2)^4 - 8(2)^2 = -16$$

$$f''(x) = 12(2^2) - 16 > 0$$

$\therefore$ MIN at $(2, -16)$

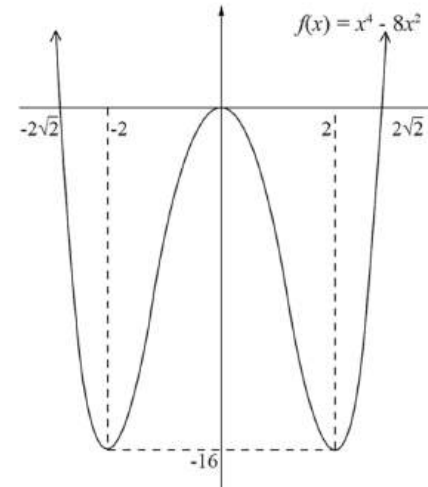
$\therefore$ MIN at $(-2, -16)$, ($f(x)$ is even)

$$\text{At } x = 0 \quad f(x) = (0)^4 - 8(0)^2 = 0$$

$$f''(x) = 12(0^2) - 16 < 0$$

$\therefore$ MAX at $(0, 0)$

iv.



i. $f(x) = \frac{\log_e x}{x}$, for $x > 0$

Using the quotient rule,

$$f'\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$$

$$f'(x) = \frac{x \cdot \frac{d}{dx}(\log_e x) - \log_e x \cdot \frac{d}{dx}(x)}{x^2}$$

$$= \frac{x\left(\frac{1}{x}\right) - \log_e x \cdot 1}{x^2}$$

$$= \frac{1 - \log_e x}{x^2}$$

SP's occur when $f'(x) = 0$

$$\frac{1 - \log_e x}{x^2} = 0$$

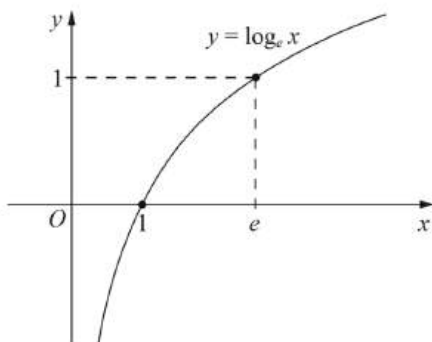
$$1 - \log_e x = 0$$

$$\log_e x = 1$$

$$\therefore x = e$$

$\therefore$ There is a stationary point at $x = e$.

ii.



When $x < e$, $\log_e x < 1$ (from graph)

$$1 - \log_e x > 0$$

$$\frac{1 - \log_e x}{x^2} > 0$$

$$\therefore f'(x) > 0$$

When $x > e$, $\log_e x > 1$ (from graph)

$$1 - \log_e x < 0$$

$$\frac{1 - \log_e x}{x^2} < 0$$

$$\therefore f'(x) < 0$$

$\therefore$ The stationary point at $x = e$ is a maximum.

iii. $f(e) = \frac{\log_e e}{e}$

$$= \frac{1}{e}$$

$\therefore f(x)$ has a maximum value at $\left(e, \frac{1}{e}\right)$

$$\therefore \frac{\log_e x}{x} \leq \frac{1}{e}$$

$$\log_e x \leq \frac{x}{e}$$

$$e \log_e x \leq x$$

$$\log_e x^e \leq x$$

$$e^{\log_e x^e} \leq e^x$$

$$x^e \leq e^x \quad (\text{using } e^{\log x} = x)$$

$$\therefore e^x \geq x^e \quad (\text{for } x > 0)$$

MARKER'S COMMENT: Very challenging for most students. Successful students recognised the link to part (ii).

a. $f(x) = x - \frac{x^2}{2} + \frac{x^3}{3}$

Turning points when $f'(x) = 0$

$$f'(x) = 1 - x + x^2$$

$$x^2 - x + 1 = 0$$

Since $\Delta = b^2 - 4ac$

$$= (-1)^2 - 4 \times 1 \times 1$$

$$= -3 < 0 \Rightarrow \text{No solution}$$

$\therefore f(x)$ has no turning points

b. P.I. when $f''(x) = 0$

$$f''(x) = -1 + 2x = 0$$

$$2x = 1$$

$$x = \frac{1}{2}$$

Check for change in concavity

$$f''\left(\frac{1}{4}\right) = -\frac{1}{2} < 0$$

$$f''\left(\frac{3}{4}\right) = \frac{1}{2} > 0$$

$\Rightarrow$ Change in concavity

$$\therefore \text{P.I. at } x = \frac{1}{2}$$

$$\begin{aligned} f\left(\frac{1}{2}\right) &= \frac{1}{2} - \frac{\left(\frac{1}{2}\right)^2}{2} + \frac{\left(\frac{1}{2}\right)^3}{3} \\ &= \frac{1}{2} - \frac{1}{8} + \frac{1}{24} \\ &= \frac{5}{12} \end{aligned}$$

$\therefore$ Point of Inflection at $\left(\frac{1}{2}, \frac{5}{12}\right)$

♦♦ Mean mark 28% for all of Q10
(note that data for each question
part is not available).

c.i. Show $1 - x + x^2 - \frac{1}{1+x} = \frac{x^3}{1+x}, \quad x \neq -1$

$$\begin{aligned} \text{LHS} &= \frac{1+x}{1+x} - \frac{x(1+x)}{1+x} + \frac{x^2(1+x)}{(1+x)} - \frac{1}{1+x} \\ &= \frac{1+x - x - x^2 + x^2 + x^3 - 1}{1+x} \\ &= \frac{x^3}{1+x} \quad \dots \text{ as required} \end{aligned}$$

c.ii. Let $g(x) = \ln(1+x)$

$$g'(x) = \frac{1}{1+x}$$

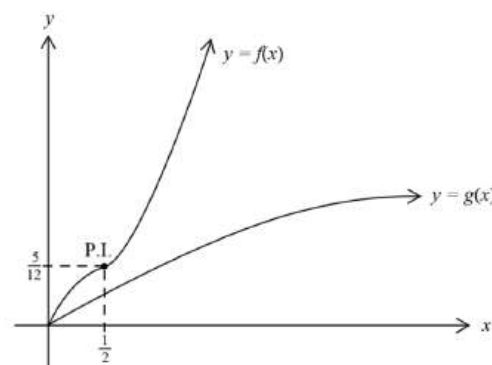
$$\begin{aligned} f'(x) - g'(x) &= 1 - x + x^2 - \frac{1}{1+x} \\ &= \frac{x^3}{1+x} \quad (\text{using part (i)}) \end{aligned}$$

Since $\frac{x^3}{1+x} \geq 0$ for $x \geq 0$

$$f'(x) - g'(x) \geq 0$$

$$f'(x) \geq g'(x) \text{ for } x \geq 0$$

d.



MARKER'S COMMENT: When 2
graphs are drawn on the same set
of axes, you **must** label them.

e. Show $\frac{d}{dx}[(1+x)\ln(1+x) - (1+x)] = \ln(1+x)$

Using $\frac{d}{dx}uv = uv' + vu'$

$$\begin{aligned}
\text{LHS} &= (1+x) \times \frac{1}{1+x} + \ln(1+x) \times 1 + -1 \\
&= 1 + \ln(1+x) - 1 \\
&= \ln(1+x) \\
&= \text{RHS} \quad \dots \text{ as required}
\end{aligned}$$

$$\begin{aligned}
\text{f. Area} &= \int_0^1 f(x) - g(x) dx \\
&= \int_0^1 \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \ln(x+1) \right) dx \\
&= \left[\frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{12} - (1+x)\ln(1+x) + (1+x) \right]_0^1 \\
&\quad \text{(using part (e) above)} \\
&= \left[\left(\frac{1}{2} - \frac{1}{6} + \frac{1}{12} - (2)\ln 2 + 2 \right) - (\ln 1 + 1) \right] \\
&= \frac{5}{12} - 2\ln 2 + 2 - 1 \\
&= 1\frac{5}{12} - 2\ln 2 \text{ u}^2
\end{aligned}$$

17. Calculus, 2ADV C3 2012 HSC 14a

$$\begin{aligned}
\text{i.} \quad f(x) &= 3x^4 + 4x^3 - 12x^2 \\
f'(x) &= 12x^3 + 12x^2 - 24x \\
f''(x) &= 36x^2 + 24x - 24
\end{aligned}$$

Stationary points when $f'(x) = 0$

$$\Rightarrow 12x^3 + 12x^2 - 24x = 0$$

$$12x(x^2 + x - 2) = 0$$

$$12x(x+2)(x-1) = 0$$

$\therefore$ Stationary points at $x = 0, 1$ or -2

$$\text{When } x = 0, \quad f(0) = 0$$

$$f''(0) = -24 < 0$$

$\therefore$ MAX at $(0,0)$

When $x = 1$

$$f(1) = 3 + 4 - 12 = -5$$

$$f''(1) = 36 + 24 - 24 = 36 > 0$$

$\therefore$ MIN at $(1,-5)$

When $x = -2$

$$f(-2) = 3(-2)^4 + 4(-2)^3 - 12(-2)^2$$

$$= 48 - 32 - 48$$

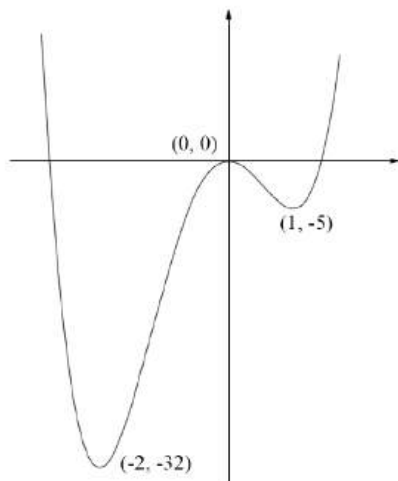
$$= -32$$

$$f''(-2) = 36(-2)^2 + 24(-2) - 24$$

$$= 144 - 48 - 24 = 72 > 0$$

$\therefore$ MIN at $(-2,-32)$

ii.



- iii. $f(x)$ is increasing for
 $-2 < x < 0$ and $x > 1$

♦ Mean mark 42%

MARKER'S COMMENT: Be careful to use the correct inequality signs, and not carelessly include $\geq$ or $\leq$ by mistake.

- iv. Find k such that

$$3x^4 + 4x^3 - 12x^2 + k = 0 \text{ has no solution}$$

k is the vertical shift of $y = 3x^4 + 4x^3 - 12x^2$

$\Rightarrow$ No solution if it does not cross the x -axis.

$\therefore$ No solution when $k > 32$

♦♦♦ Mean mark 12%